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Lecture 25 3/25/26

Fourier Series for the Driven Oscillator

EOM: $\ddot{x} + 2\beta\dot{x} + \omega_0^2 x = f$ $\qquad Dx = f$, $D = \frac{d^2}{dt^2} + 2\beta\frac{d}{dt} + \omega_0^2$

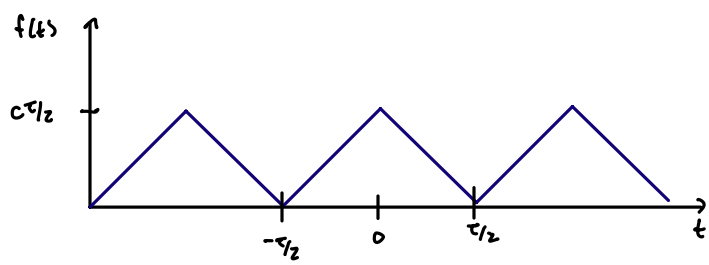
$f(t) = \sum_n f_n(t)$ $\qquad$ let $\qquad x = \sum_n x_n(t)$

$$Dx = D\left(\sum_n x_n(t)\right) = \sum_n Dx_n(t) = \sum_n f_n(t) = f(t)$$

Then $Dx = f$ $\Rightarrow$ x is a solution!

1) Find f_n
2) Find A_n, δ_n $\Rightarrow x_n(t)$
3) Write down $x(t) = \sum_n x_n(t)$

Example:



rise $= \pm c\tau/2$
run $= \tau/2$ $\Rightarrow$ slope $= \pm c$

$$f(t) = \sum_n a_n \cos n\omega t + b_n \sin n\omega t$$

$$a_0 = \frac{1}{\tau}\int_{-\tau/2}^{\tau/2} f(t)\,dt \qquad f(t) = \begin{cases} ct + c\tau/2 & -\tau/2 < t < 0 \\ -ct + c\tau/2 & 0 < t < \tau/2 \end{cases}$$

This function is even $\Rightarrow b_n = 0$

$$= \frac{1}{\tau}\int_{-\tau/2}^{0}(ct + c\tau/2)\,dt + \frac{1}{\tau}\int_0^{\tau/2}(-ct + c\tau/2)\,dt$$

$$= \frac{c}{\tau}\left[\frac{1}{2}t^2 + \tau/2\, t\right]_{-\tau/2}^{0} + \frac{c}{\tau}\left[-\frac{1}{2}t^2 + \tau/2\, t\right]_0^{\tau/2}$$

$$= \frac{c}{\tau}\left[-\frac{1}{2}\tau^2/4 + \tau/2\,\tau/2\right] + \frac{c}{\tau}\left[-\frac{1}{2}\tau/4 + \tau/2\,\tau/2\right]$$

$$= \frac{2c}{\tau}\left[-\tau^2/8 + \tau^2/4\right] = \frac{2c}{\tau}\left[\tau^2/8\right] = c\tau/4$$

$$a_n = \frac{2}{\tau}\int_{-\tau/2}^{\tau/2} f(t)\cos n\omega t\,dt$$

$$= \frac{2}{\tau}\int_{-\tau/2}^{0} c(t + \tau/2)\cos n\omega t\,dt + \frac{2}{\tau}\int_0^{\tau/2} c(-t + \tau/2)\cos n\omega t\,dt$$

$$= \frac{2c}{\tau}\left[\int_{-\tau/2}^{0} t\cos n\omega t + \tau/2\int_{-\tau/2}^{0}\cos n\omega t\,dt - \int_{0}^{\tau/2} t\cos n\omega t + \tau/2\int_{0}^{\tau/2}\cos n\omega t\,dt\right]$$

$$\int x\cos ax = \frac{1}{a^2}\cos ax + \frac{x}{a}\sin ax$$

$$= \frac{2c}{\tau}\left[\left(\frac{1}{n^2\omega^2}\cos n\omega t + \frac{t}{n\omega}\sin n\omega t\right)_{-\tau/2}^{0} + \tau/2\left(\frac{1}{n\omega}\sin n\omega t\right)_{-\tau/2}^{0} - \left(\frac{1}{n^2\omega^2}\cos n\omega t + \frac{t}{n\omega}\sin n\omega t\right)_{0}^{\tau/2} + \tau/2\left(\frac{1}{n\omega}\sin n\omega t\right)_{0}^{\tau/2}\right]$$

$$= \frac{2c}{\tau}\left[\frac{1}{n^2\omega^2}(1) + 0 - \frac{1}{n^2\omega^2}\cos\left(\frac{n\omega\tau}{2}\right) - \left(\frac{-\tau}{2n\omega}\right)\sin\left(-\frac{n\omega\tau}{2}\right) + 0 - \tau/2\left(\frac{1}{n\omega}\right)\sin\left(-\frac{n\omega\tau}{2}\right) - \left(\frac{1}{n^2\omega^2}\cos\left(\frac{n\omega\tau}{2}\right) + \frac{\tau}{2n\omega}\sin\left(\frac{n\omega\tau}{2}\right) - \frac{1}{n^2\omega^2} - 0\right) + \tau/2\left(\frac{1}{n\omega}\sin\frac{n\omega\tau}{2}\right)\right]$$

$$\omega = \frac{2\pi}{\tau} \qquad \frac{n\omega\tau}{2} = \frac{2\pi}{\tau}\frac{n\tau}{2} = \pi n$$

$$= \frac{2c}{\tau}\left[\frac{1}{n^2\omega^2} - \frac{1}{n^2\omega^2}\cos n\pi - \cancelto{0}{\frac{\tau}{2n\omega}\sin n\pi} + \cancelto{0}{\frac{\tau}{2n\omega}\sin n\pi} - \frac{1}{n^2\omega^2}\cos n\pi - \cancelto{0}{\frac{\tau}{2n\omega}\sin n\pi} + \frac{1}{n^2\omega^2} + \cancelto{0}{\frac{\tau}{2n\omega}\sin n\pi}\right]$$

$$= \frac{2c}{\tau}\left[\frac{2}{n^2\omega^2} - \frac{2}{n^2\omega^2}\cos n\pi\right]$$

$$= \frac{2c}{\tau}\left(\frac{2}{n^2\omega^2}\right)(1-\cos n\pi)$$

$$= \frac{2c}{\tau}\left[\frac{2\tau}{n^2\omega\, 2\pi}\right](1-\cos n\pi)$$

$$= \frac{2c}{\pi n^2\omega}(1-\cos n\pi) \qquad \cos(n\pi) = (-1)^n$$

$$= \frac{2c}{\pi n^2\omega}\left[1-(-1)^n\right]$$

$$a_n = \begin{cases} \dfrac{4c}{\pi n^2\omega} & n = \text{odd} \\ 0 & n = \text{even} \end{cases}$$

$$\Rightarrow f(t) = \sum_n a_n\cos(n\omega t)$$

Recall: When $f(t) = f_0\cos\omega t$

$x_p(t) = A\cos(\omega t+\delta)$, $A = \dfrac{f_0^2}{(\omega_0^2-\omega^2)^2 + 4\beta^2\omega^2}$, $\delta = \tan^{-1}\left(\dfrac{2\beta\omega_0}{\omega_0^2-\omega^2}\right)$

Now $f(t) = \sum_n f_n(t)$ where $f_n(t) = a_n\cos(n\omega t)$

so $f_0 \to a_n$

$\omega \to n\omega$

$\Rightarrow x_n(t) = A_n\cos(n\omega t + \delta_n)$ $\quad A_n = \dfrac{a_n^2}{(\omega_0^2 - n^2\omega^2)^2 + 4\beta^2 n^2\omega^2}$, $\delta_n = \tan^{-1}\left(\dfrac{2\beta n\omega}{\omega_0^2 - n^2\omega^2}\right)$

In our case

$$A_n = \begin{cases} \dfrac{\frac{16c^2}{\pi^2 n^4 \omega^2}}{(\omega_0^2 - n^2\omega^2) + 4\beta^2 n^2 \omega^2} & n = \text{odd} \\ 0 & n = \text{even} \end{cases} \quad \& \quad \delta_n = \tan^{-1}\left(\frac{2\beta n\omega}{\omega_0^2 - n^2\omega^2}\right)$$

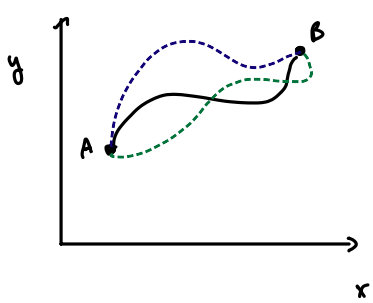
$$x(t) = \sum_{n=0}^{\infty} A_n \cos(n\omega t - \delta_n)$$

Since $A_n = 0 \quad \forall \quad n = \text{even}$ call $2k+1 = n$

$$x(t) = \sum_{k=0} A_k \cos(k\omega t - \delta_k)$$

$$\delta_k = \tan^{-1}\left(\frac{2\beta(2k+1)\omega}{\omega_0^2 - (2k+1)^2\omega^2}\right) \qquad A_k^2 = \frac{16c^2}{\pi^2(2k+1)^4\omega^2\left[\{\omega_0^2 - (2k+1)^2\omega^2\}^2 + 4(2k+1)^2\omega^2\right]}$$

Chapter 6: Calculus of Variations

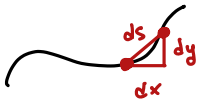


How can we find the "right" path?

Example: path that minimizes the distance between A & B

$$L = \int ds$$

$$ds^2 = d\vec{r} \cdot d\vec{r} \Rightarrow ds = \sqrt{dx^2 + dy^2}$$



$$ds = \sqrt{(dx)^2\left(1 + \frac{dy^2}{dx^2}\right)}$$

$$= dx\sqrt{1 + \left(\frac{dy}{dx}\right)^2}$$

$$ds = \sqrt{1 + y'(x)^2}\, dx$$

$$\Rightarrow L = \int \sqrt{1 + y'(x)^2}\, dx$$